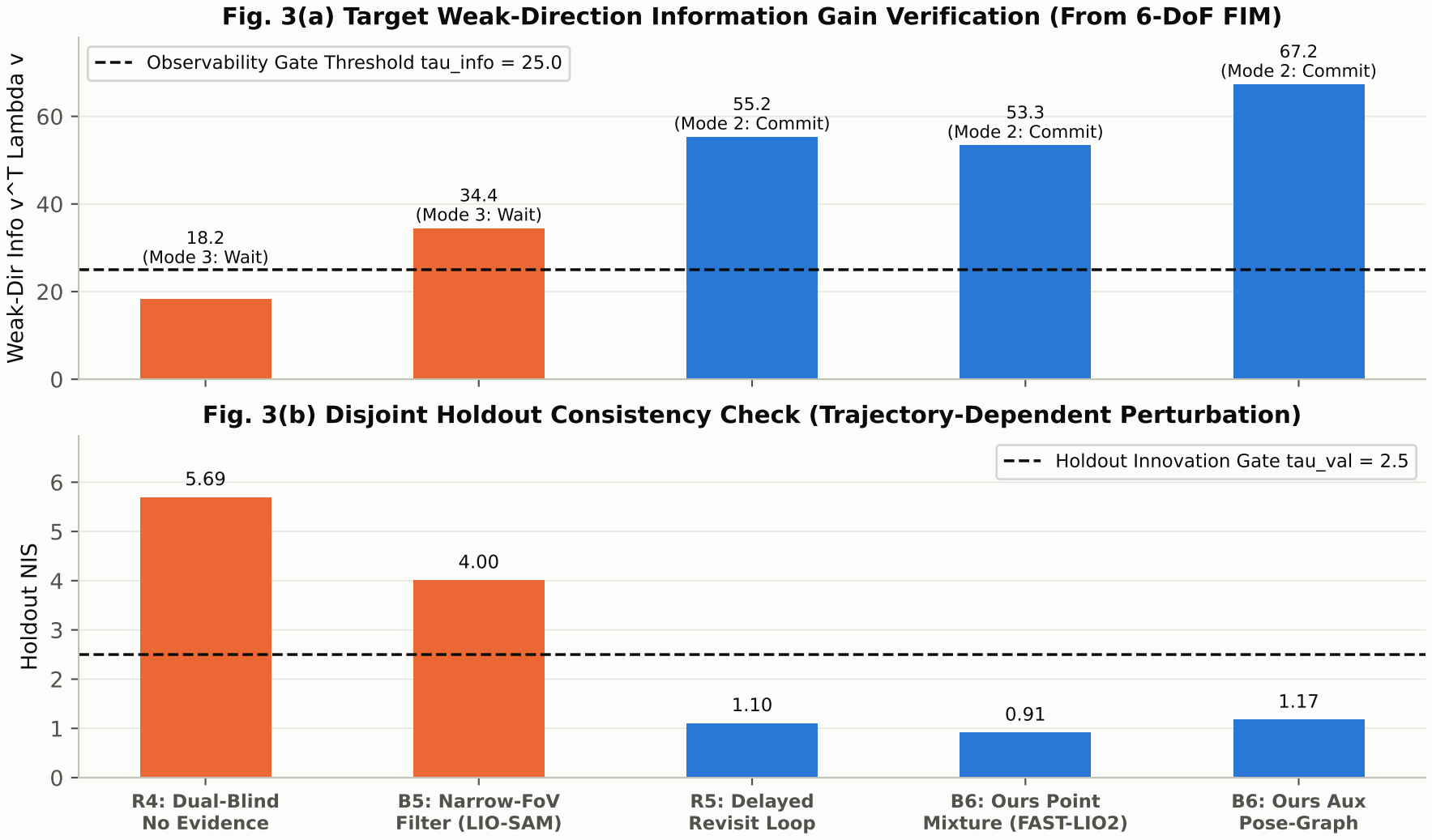
Fig. 3(a) Target Weak-Direction Information Gain Verification (From 6-DoF FIM)
Observability Gate Threshold tau_info = 25.0
Weak-Dir Info v^T Lambda v
0
20
40
60
18.2
(Mode 3: Wait)
34.4
(Mode 3: Wait)
55.2
(Mode 2: Commit)
53.3
(Mode 2: Commit)
67.2
(Mode 2: Commit)
Fig. 3(b) Disjoint Holdout Consistency Check (Trajectory-Dependent Perturbation)
Holdout Innovation Gate tau_val = 2.5
Holdout NIS
0
1
2
3
4
5
6
5.69
4.00
1.10
0.91
1.17
R4: Dual-Blind
No Evidence
B5: Narrow-FoV
Filter (LIO-SAM)
R5: Delayed
Revisit Loop
B6: Ours Point
Mixture (FAST-LIO2)
B6: Ours Aux
Pose-Graph